Channel, sediment, nutrient

Purpose

SWAT+ supports two channel-routing implementations, each with its own output set.

- **Legacy channel** (`channel_*.txt`, `basin_cha_*.txt`). One row per channel object, holding inflow / outflow of water, sediment, sediment by particle size, nutrients, pesticides, bacteria, and conservative metals.
- **sd_channel** (`channel_sd_*.txt`, `basin_sd_cha_*.txt`). Storage-and-discharge routing. Three blocks of columns: a `water_body` block (area, precip, evap, seep, storage), followed by an inflow `hyd_output` record and an outflow `hyd_output` record.

The channel sediment budget and morphology have their own writers (`sd_chanbud_output.f90`, `sd_chanmorph_output.f90`) and produce `*_chanbud_*.txt` and `*_chamorph_*.txt`.

Legacy channel files

Source

- Writer: src/channel_output.f90, units 2480 (day), 2481 (mon), 2482 (yr), 2483 (aa).
- Basin sum: src/basin_channel_output.f90.
- File opening: src/header_channel.f90.
- Record type: src/channel_module.f90, `type ch_output`.

Format

Each line is the seven leading columns (jday, mon, day, yr, unit, gis_id, name), then the 60 fields of `ch_output` in declaration order.

#	Column	Units	Description
1	flo_in	ha-m	streamflow into reach
2	flo_out	ha-m	streamflow out of reach
3	evap	m^3/s	rate of water loss by evaporation
4	tloss	m^3/s	rate of transmission loss through streambed
5	sed_in	tons	sediment transported into reach
6	sed_out	tons	sediment transported out of reach
7	sed_conc	mg/L	sediment concentration in reach
8	orgn_in	kg N	organic nitrogen in
9	orgn_out	kg N	organic nitrogen out
10	orgp_in	kg P	organic phosphorus in
11	orgp_out	kg P	organic phosphorus out
12	no3_in	kg N	nitrate in
13	no3_out	kg N	nitrate out
14	nh4_in	kg	ammonium in
15	nh4_out	kg	ammonium out
16	no2_in	kg	nitrite in
17	no2_out	kg	nitrite out
18	solp_in	kg P	soluble phosphorus in
19	solp_out	kg P	soluble phosphorus out
20	chla_in	kg	chlorophyll-a in
21	chla_out	kg	chlorophyll-a out
22	cbod_in	kg	carbonaceous BOD in

#	Column	Units	Description
23	cbod_out	kg	carbonaceous BOD out
24	dis_in	kg	dissolved oxygen in
25	dis_out	kg	dissolved oxygen out
26	solpst_in	mg pst	soluble pesticide in
27	solpst_out	mg pst	soluble pesticide out
28	sorbpst_in	mg pst	sorbed pesticide in
29	sorbpst_out	mg pst	sorbed pesticide out
30	react	mg pst	pesticide loss by reaction in water
31	volat	mg	pesticide loss by volatilization
32	setlpst	mg pst	pesticide settling from water to bed
33	resuspst	mg	pesticide resuspended from bed
34	difus	mg	pesticide diffused from water to bed
35	reactb	mg	pesticide reaction loss in bed
36	bury	mg	pesticide buried
37	sedpest	mg	pesticide in bed sediment
38	bacp	# cfu/100mL	persistent bacteria out
39	baclp	# cfu/100mL	less persistent bacteria out
40	met1	kg	conservative metal #1 out
41	met2	kg	conservative metal #2 out
42	met3	kg	conservative metal #3 out
43	sand_in	tons	sand in
44	sand_out	tons	sand out
45	silt_in	tons	silt in
46	silt_out	tons	silt out
47	clay_in	tons	clay in
48	clay_out	tons	clay out
49	smag_in	tons	small aggregate in
50	smag_out	tons	small aggregate out
51	lag_in	tons	large aggregate in
52	lag_out	tons	large aggregate out
53	grvl_in	tons	gravel in
54	grvl_out	tons	gravel out
55	bnk_ero	tons	bank erosion
56	ch_deg	tons	channel degradation
57	ch_dep	tons	channel deposition
58	fp_dep	tons	floodplain deposition
59	tot_ssed	mg/L	total suspended sediments

Example header

`refdata/Ames_sub1/basin_cha_aa.txt`:

```
 demo                        SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328
  jday     mon    day    yr     unit  gis_id  name                  flo_in        flo_out             evap
                                                                      ha-m           ha-m             ha-m
```

The header reports `evap` and `tloss` units as `ha-m`, but `ch_output` carries them as `m^3/s` (rate). The unit row in the file does not match the source comment.

sd_channel files

Source

- Writer: src/sd_channel_output.f90, units 2500 (day), 2501 (mon), 2502 (yr), 2503 (aa); subdaily on unit 2508.
- Basin sum: src/basin_sdchannel_output.f90.
- File opening: src/header_sd_channel.f90.
- Record types: src/water_body_module.f90, `type water_body`, and src/hydrograph_module.f90, `type hyd_output`.

Format

Each record is the seven leading columns, then four blocks of fields written in this order by `sd_channel_output.f90`:

```
write (2500,100) time%day, time%mo, time%day_mo, time%yrc, ichan, ob(iob)%gis_id, ob(iob)%name,
 ↪  &
  ch_wat_d(ichan)%area_ha, ch_wat_d(ichan)%precip, ch_wat_d(ichan)%evap, ch_wat_d(ichan)%seep,
 ↪  &
  ch_stor(ichan), ch_in_d(ichan), ch_out_d(ichan), wtemp
```

Block 1: water body status (4 fields).

#	Column	Units	Description
1	area	ha	channel water surface area
2	precip	m^3	precipitation on water surface
3	evap	m^3	evaporation from water surface
4	seep	m^3	seepage from streambed

Block 2: channel storage `ch_stor` (one `hyd_output` record, 18 fields). These are the `_stor` columns in the file header.

Block 3: inflow `ch_in_d` (one `hyd_output` record, 18 fields). These are the `_in` columns.

Block 4: outflow `ch_out_d` (one `hyd_output` record, 18 fields). These are the `_out` columns.

Block 5: water temperature `wtemp` (deg C), one trailing scalar.

The 18 fields of `hyd_output`, in declaration order:

Field	Units	Description
flo	m^3 (storage), m^3/s (in / out flow rate)	volume or flow rate
sed	metric tons	sediment
orgn	kg N	organic N
sedp	kg P	organic P
no3	kg N	NO3-N
solp	kg P	soluble P
chla	kg	chlorophyll-a
nh3	kg N	NH3
no2	kg N	NO2
cbod	kg	carbonaceous BOD
dox	kg	dissolved oxygen
san	tons	detached sand
sil	tons	detached silt
cla	tons	detached clay
sag	tons	detached small aggregate
lag	tons	detached large aggregate
grv	tons	gravel
temp	deg C	temperature

The line ends with `wtemp` after the `out` block.

Example header

`refdata/Ames_sub1/basin_sd_cha_aa.txt`:

```
 demo                      SWAT+ 2024-08-27        MODULAR Rev 2024.61.0.1-33-g68b1328
   jday    mon    day     yr      unit   gis_id   name                    area          precip           evap
                                                                            ha             m^3            m^3
```

Each block ends with a `null`-labelled column. That column is the `temp` field of `hyd_output`. The writer passes `ch_in_d(ichan)`, `ch_out_d(ichan)`, and `ch_stor(ichan)` as whole records, so the 18th field of each block is `temp`. The basin-wide water temperature is the trailing `wtemp` scalar written after the outflow block.

Channel sediment budget and morphology

- `sd_chanbud_*.txt`. Sediment budget per channel: bank erosion, bed erosion, deposition. Writer: `src/sd_chanbud_output.f90`. Basin sum in `src/basin_chanbud_output.f90`.
- `sd_chamorph_*.txt`. Channel morphology: width, depth, slope. Writer: `src/sd_chanmorph_output.f90`. Basin sum in `src/basin_chanmorph_output.f90`.

Related files

- Aquifer, reservoir, plant outputs.
- Water balance outputs.